

Red Team Agent – Lab Walkthrough

OWASP Juice Shop | Local LLM (llama3.2) | Kali Linux

Target: http://localhost:8082

Overview

This walkthrough documents the exact steps taken to get the `run_agent.py` red team script working end-to-end in the lab environment. Follow each step in order. Screenshots should be inserted where placeholders appear.

Step 1 – Start OWASP Juice Shop

The target application must be running on port 8082 before the scans can return meaningful results. Pull and start the Juice Shop Docker container.

```
docker run -d -p 8082:3000 --name juice-shop bkimminich/juice-shop
```

```
(kali@kali)-[~/Labfiles5.1]
$ docker run -d -p 8082:3000 --name juice-shop bkimminich/juice-shop
```

Wait approximately 30 seconds for the container to start, then verify it is up:

```
docker ps | grep juice-shop
```

```
curl http://localhost:8082
```

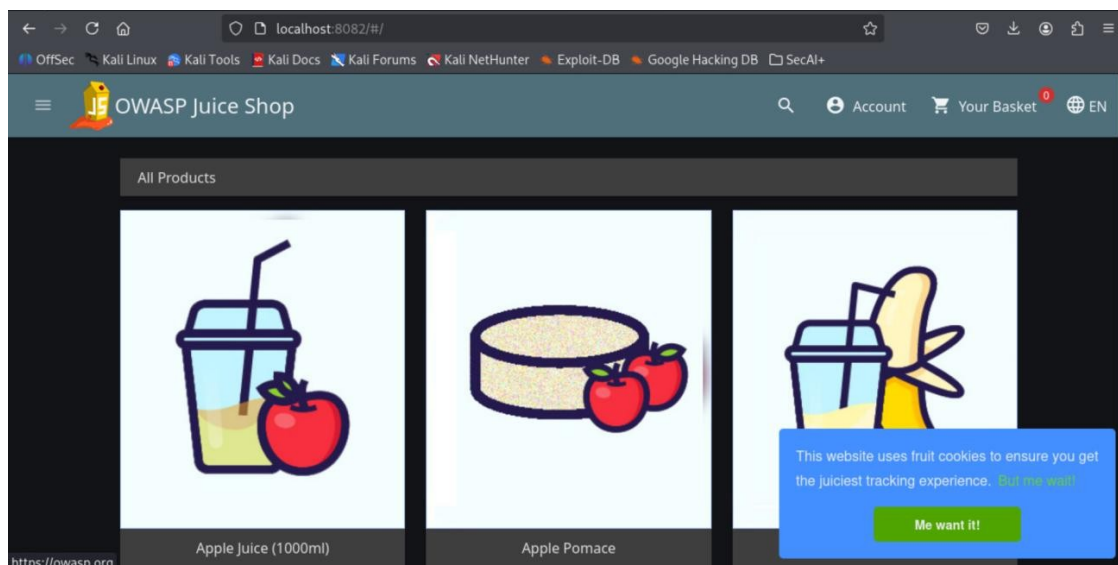
```
(kali@kali) ~/[Labfiles 5.1]
└─$ curl http://localhost:8082
```

```
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- SPDX-License-Identifier: MIT
```

```
<!doctype html>
<html lang="en" data-beasties-container>
<head>
  <meta charset="utf-8">
  <title>OWASP Juice Shop</title>
  <meta name="description" content="Probably the most modern and sophisticated insecure web application">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="preconnect" href="https://fonts.googleapis.com/">
  <link rel="preconnect" href="https://fonts.gstatic.com/" crossorigin>
  <script>font-face{font-family:'VT323';font-style:normal;font-weight:400;font-display:swap;src:url(https://fonts.gstatic.com/s/vt323/v18/pkxipyp0ihIEF2isQfJXgdd,woff2) format('woff2');unicode-range:U+0102-0103,U+0110-0111,U+0128-0129,U+0168-0169,U+01A0-01A1,U+01AF-01B0,U+0300-0301,U+0303-0304,U+0308-0309,U+0323,U+0329,U+039A-039F,U+04AB-04AF;}font-face{font-family:'VT323';font-style:normal;font-weight:400;font-display:swap;src:url(https://fonts.gstatic.com/s/vt323/v18/pkxipyp0ihIEF2isRFJXgdd,woff2) format('woff2');unicode-range:U+0100-02BA,U+02C7-02C8,U+02C9-02CA,U+02CE-02D7,U+02DD-02FF,U+0304,U+0308-0329,U+039A-039F,U+04AB-04AF,U+04E0-04E9,U+04F2-04FF,U+0500-050A,U+05AD-05AE,U+05B0-05B9,U+05C0-05C9,U+05D0-05D9,U+05E0-05E9,U+05F0-05F9,U+0600-060F,U+0610-061F,U+0620-0629,U+0630-0639,U+0640-0649,U+0650-0659,U+0660-066F,U+0670-067F,U+0680-068F,U+0690-069F,U+06A0-06A9,U+06B0-06B9,U+06C0-06C9,U+06D0-06D9,U+06E0-06EF,U+06F0-06FF};font-family:'VT323';font-style:normal;font-weight:400;font-display:swap;src:url(https://fonts.gstatic.com/s/vt323/v18/pkxipyp0ihIEF2isfJ3U,woff2) 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```

If you get such HTML script response it means it is up

You can also open `http://localhost:8082` in the browser to confirm the Juice Shop UI loads.



Step 2 – Final Successful Run

With Juice Shop running and agent_local.py updated, run the script again. All six tools will execute against the live target and the LLM will generate a security report.

```
cd ~/Labfiles5.1
python3 run_agent.py
```

```
(kali㉿kali)-[~/Labfiles5.1]
$ python3 run_agent.py
```

The script will print progress for each tool as it runs:

- scan_website – checks HTTP status and headers
- discover_api_endpoints – probes /api/, /rest/, /swagger.json, etc.
- enumerate_directories – checks /admin, /login, /robots.txt, /ftp, etc.
- check_admin_endpoints – checks admin-specific REST paths
- juiceshop_xss_search_alert – tests XSS via the search endpoint
- juiceshop_sql_i_admin_login_bypass – tests SQL injection on the login endpoint

After the tools finish the LLM analyses the results and writes the report. Expected findings:

- SQLi: vulnerable = true (admin login bypass confirmed)
- XSS: payload reflected via search route
- Multiple admin endpoints returning HTTP 200

Step 3 – View the Security Report

The report is saved as security_report_TIMESTAMP.txt in the Labfiles5.1 directory. Open it to review the full findings.

```
cat ~/Labfiles5.1/security_report_*.txt
```

```
(kali㉿kali)-[~/Labfiles5.1]
$ cat ~/Labfiles5.1/security_report_*.txt
```

The report contains: vulnerability summary, accessible paths, raw tool output, AI analysis section, and a high-level AI synopsis (3-5 bullet points suitable for training use).
